

MOS-Based Low-Dose CT Image Quality Assessment with Vision-Language Models

Author Name
Department or Laboratory Name
Institution Name
author.email@example.com

Abstract

Low-dose computed tomography (LDCT) reduces radiation exposure but introduces image degradation that can affect diagnostic confidence. This report presents a mean-opinion-score (MOS) based image quality assessment pipeline for LDCT images using vision-language models (VLMs). The repository implements preprocessing from TIFF to normalized PNG and JSONL data, supervised fine-tuning (SFT) with LoRA adapters, optional MOS-regression training, GRPO-style refinement, and standardized evaluation. Final experiments include six SFT and seven GRPO configurations on 300 test samples. The selected SFT sweep run reaches SROCC 0.932 with complete prediction coverage, while the selected GRPO run reaches SROCC 0.883.

Keywords: low-dose CT, image quality assessment, MOS prediction, vision-language model, supervised fine-tuning

1. Introduction

Low-dose CT acquisition is clinically attractive because it can lower patient radiation exposure. The cost is increased noise, altered texture, and possible loss of diagnostically useful detail. Automatic image quality assessment (IQA) can support model selection, reconstruction tuning, and quality-control workflows by estimating the subjective quality of a scan from image evidence.

The repository accompanying this report targets MOS-based LDCT IQA. It trains and evaluates VLM-based predictors that receive a CT image and produce a scalar quality score. The central design goal is practical reproducibility: the code separates preprocessing, dataset conversion, training, evaluation, and model comparison into reusable scripts and JSON configuration files.

The main contribution of this report is a compact paper-style description of the implemented pipeline and the final experimental artifacts. We summarize the data flow, training modes, evaluation metrics, and observed performance of the base VLM, SFT, and GRPO checkpoints.

2. Related Work

Classical full-reference IQA metrics such as SSIM model perceptual fidelity through structural image similarity rather than only pixel error [1]. Statistical image information has also been used to connect visual fidelity with natural image regularities [2]. More recent perceptual metrics use deep feature spaces to align better with human judgments [3].

For this project, MOS prediction is treated as an image-conditioned language task. A medical VLM is adapted using

lightweight LoRA modules [4], allowing the base model to remain frozen while task-specific parameters are trained. The repository also includes configuration support for TRL-style SFT and GRPO workflows [5], with MedGemma used as the base model family [6].

3. Dataset and Preprocessing

The expected raw data layout contains TIFF LDCT images and MOS annotations. The preprocessing entrypoint converts source scans into model-ready PNG images and writes split-specific JSONL files:

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N, \quad (1)$$

where x_i is an LDCT image and y_i is its MOS target. The preprocessing configuration supports min-max or percentile normalization and optional resizing.

After preprocessing, the repository stores base JSONL files under `datasets/processed/`. For TRL training, JSONL files can be converted into Arrow datasets using `scripts/build_sft_arrow_dataset.py`. This separation keeps data normalization independent from the training format required by a specific trainer.

4. Method

4.1. Prompted MOS Prediction

The base VLM is evaluated by prompting it to predict the MOS score directly from an image. This establishes the zero- or few-instruction baseline for the same downstream parser and evaluation code used by fine-tuned models.

4.2. Supervised Fine-Tuning

The SFT path uses image-chat examples in which the model is instructed to return the MOS score. LoRA adapters are trained on top of the base VLM, reducing trainable parameters while preserving the general medical vision-language representation.

At inference time, generated text is parsed into a scalar score. Invalid generations are tracked separately, making the prediction rate a first-class evaluation statistic:

$$r_{\text{valid}} = \frac{N_{\text{valid}}}{N_{\text{total}}}. \quad (2)$$

4.3. Regression and GRPO Modes

The codebase also contains a MOS regression trainer and GRPO refinement path. Regression mode predicts MOS through a direct scoring head. GRPO mode can start from an SFT LoRA adapter or directly from the base VLM. The

Table 1: Selected test-set MOS prediction results. Lower MAE/RMSE is better; higher correlation and coverage are better. Sweep selections are based on maximum SROCC.

Model	MAE	RMSE	PLCC	SROCC	KROCC	Coverage
Direct base VLM	4.453	5.395	0.014	-0.020	-0.016	100.0%
Standalone SFT	0.347	0.451	0.916	0.917	0.787	100.0%
Selected SFT	0.393	0.482	0.930	0.932	0.808	100.0%
Selected GRPO	0.654	0.790	0.859	0.883	0.764	100.0%

final study uses four generated completions and rewards MOS closeness with negative absolute error:

$$R(\hat{y}, y) = -|\hat{y} - y|. \quad (3)$$

The GRPO study varies learning rate, temperature, KL coefficient, completion length, and initialization. These modes are available through the shared `scripts/train.py` and JSON configuration interface.

5. Evaluation

Evaluation is performed on the held-out test split. Let $\hat{y}_i$ denote the predicted MOS and y_i the ground truth MOS. The primary error metrics are

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i| \quad (4)$$

and

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2}. \quad (5)$$

Correlation is reported with PLCC, SROCC, and KROCC to capture linear agreement, monotonic ranking, and pairwise ranking consistency.

5.1. Selected Models

The sweep scripts select runs by maximum SROCC. Table 1 compares the direct baseline, the standalone SFT adapter used to initialize SFT-based GRPO runs, and the selected runs from the final sweeps.

Supervised adaptation produces the strongest results. The standalone SFT adapter has the lowest absolute errors, while the selected low-rank SFT sweep run has the strongest correlations. The selected GRPO run is the best GRPO configuration by SROCC and preserves complete prediction coverage, but it does not improve on the SFT results.

5.2. SFT Parameter Study

The SFT study evaluates epoch count, learning rate, LoRA rank, and dropout. All six configurations produce valid scores for all 300 test samples. As shown in Table 2, the selected configuration uses two epochs, learning rate 2×10^{-4} , rank 8 LoRA adapters, and dropout 0.05. The three-epoch baseline has slightly lower error and higher PLCC, but the low-rank run has the best SROCC and KROCC.

Table 2: Final SFT parameter study. The selected run is bold. All runs have 100% prediction coverage.

SFT configuration	MAE	PLCC	SROCC	KROCC
Baseline, 1 epoch	0.360	0.919	0.922	0.794
Baseline, 2 epochs	0.384	0.930	0.931	0.804
Baseline, 3 epochs	0.368	0.933	0.932	0.807
Lower LR, 1×10^{-4}	0.374	0.931	0.932	0.804
Low-rank LoRA, rank 8	0.393	0.930	0.932	0.808
Higher dropout, 0.10	0.376	0.928	0.929	0.804

Table 3: Final GRPO parameter study. The selected run is bold. “SFT” denotes initialization from the standalone SFT adapter.

GRPO configuration	MAE	PLCC	SROCC	KROCC	Coverage
SFT init., baseline	0.690	0.854	0.883	0.763	99.3%
SFT init., KL 0.02	0.654	0.854	0.877	0.759	97.0%
SFT init., LR 5×10^{-6}	0.654	0.859	0.883	0.764	100.0%
SFT init., temperature 1.0	0.795	0.762	0.814	0.705	55.3%
SFT init., length 48	0.683	0.869	0.881	0.764	100.0%
Base init., LR 1×10^{-5}	0.974	0.076	0.073	0.061	89.3%
Base init., LR 5×10^{-6}	1.027	-0.042	-0.042	-0.035	89.3%

5.3. GRPO Parameter Study

The GRPO study contains five SFT-initialized configurations and two runs initialized directly from the base VLM. Table 3 shows that the lower learning rate of 5×10^{-6} gives the strongest GRPO SROCC while retaining complete coverage. Higher temperature sharply reduces the valid prediction rate. Direct base-model initialization also performs poorly, indicating that the SFT starting point is important for this reward-based refinement stage.

6. Implementation

The project is organized around reusable command-line entrypoints. Preprocessing is handled by `scripts/preprocess_ldct.py`. Training is launched through `scripts/train.py`, with the selected mode controlled by JSON files such as `config/sft.json`, `config/grpo.json`, and `config/regression.json`. Evaluation is run with `scripts/evaluate.py`, and model comparisons are generated with `scripts/compare_models.py`.

This configuration-first layout makes the experiments easy to rerun on an HPC system. Shell scripts such as `run.sh` and `run_al00.sh` can wrap the same Python entrypoints while preserving the model, data, and output paths in versioned JSON files.

7. Discussion

The final results indicate that MOS scoring benefits strongly from task-specific supervision. The SFT variants consistently achieve correlations above 0.91 while preserving complete prediction coverage. The selected low-rank adapter is especially useful because it reaches the highest SROCC with a smaller LoRA rank.

GRPO behavior is more sensitive to initialization and generation settings. Starting from the SFT adapter produces useful rank correlations, whereas starting directly from the base

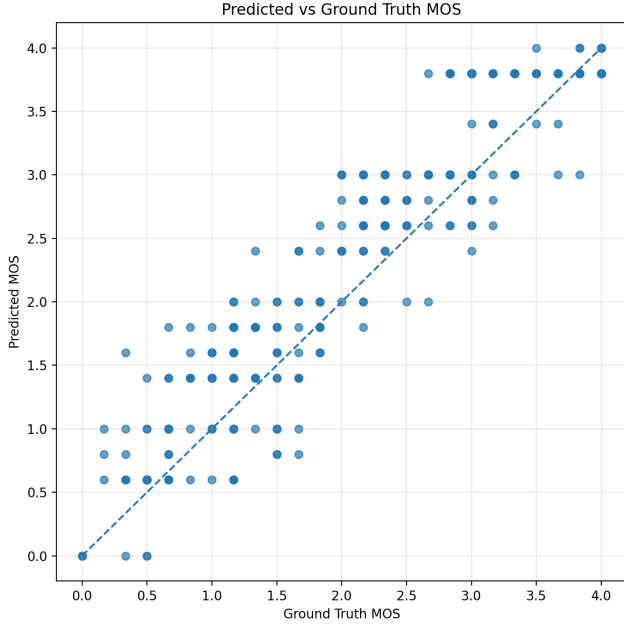


Figure 1: Predicted versus ground-truth MOS for the selected SFT run. The dashed line denotes ideal agreement.

VLM does not. The selected GRPO run remains weaker than SFT, and its discrete prediction bands in Figure 2 suggest that reward refinement compresses the predicted score range. Future experiments should evaluate regression checkpoints, add confidence intervals across seeds, and analyze errors by CT anatomy, dose level, and MOS range.

8. Conclusion

This report documents a paper-style LDCT IQA pipeline based on MOS supervision and VLM adaptation. The final parameter studies show that supervised adaptation is the strongest tested approach: the selected low-rank SFT run reaches SROCC 0.932 with complete prediction coverage. GRPO remains viable when initialized from SFT, but the tested refinements do not surpass the supervised models. The repository structure supports further experiments with regression heads, SFT adapters, and reward-based refinement using the same preprocessing and evaluation interfaces.

References

- [1] Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004.
- [2] Hamid R. Sheikh and Alan C. Bovik. Image information and visual quality. *IEEE Transactions on Image Processing*, 15(2):430–444, 2006.
- [3] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. *Proceedings of*

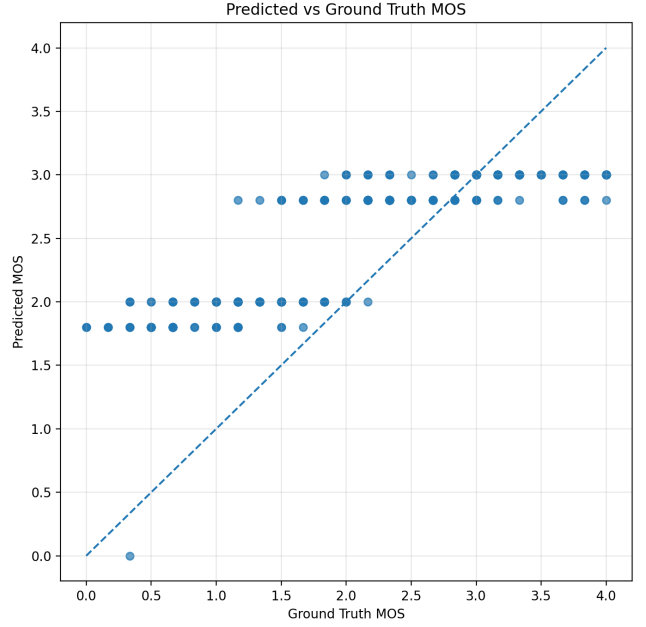


Figure 2: Predicted versus ground-truth MOS for the selected GRPO run. The discrete prediction bands indicate range compression after refinement.

the IEEE Conference on Computer Vision and Pattern Recognition, pages 586–595, 2018.

- [4] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. *International Conference on Learning Representations*, 2022.
- [5] Hugging Face. TRL: Transformer reinforcement learning, 2025. Training toolkit for supervised fine-tuning and preference optimization.
- [6] Google. MedGemma: Open medical vision-language models, 2025. Model family and model card documentation.